



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Time, Currency & Conversions

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- * Time
- * Money Calculations
- * Exchange Rates

Time



Your notes

Time

How do I convert between different units of time?

- Time has a **number of different conversions** and does not use the decimal number system (based on 10s, 100s, etc)
 - You need to know the follow time conversions

TIME CONVERSIONS

THERE ARE... ... 60 SECONDS IN A MINUTE
... 60 MINUTES IN AN HOUR
... 24 HOURS IN A DAY
... 7 DAYS IN A WEEK
... 365 DAYS IN A YEAR
 (366 IN A LEAP YEAR)
... 52 WEEKS IN A YEAR
 (PLUS ONE DAY)
... 12 MONTHS IN A YEAR
... 1000 YEARS IN A MILLENIUM

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- You should also know the **number of days** in each **calendar month**



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DO LEARN THESE – AND HOW MANY
DAYS ARE IN EACH MONTH...

THIRTY DAYS HATH SEPTEMBER, APRIL, JUNE AND NOVEMBER,
ALL THE REST HAVE THIRTY-ONE.
EXCEPTING FEBRUARY ALONE.
AND THAT HAS TWENTY-EIGHT DAYS CLEAR,
AND TWENTY-NINE IN EACH LEAP YEAR.

JANUARY	31 DAYS
FEBRUARY	28 (29 IN A LEAP YEAR)
MARCH	31
APRIL	30
MAY	31
JUNE	30
JULY	31
AUGUST	31
SEPTEMBER	30
OCTOBER	31
NOVEMBER	30
DECEMBER	31

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What is the difference between a 12-hour clock and a 24-hour clock?

- A **12-hour clock** goes around once for AM and once for PM in **one complete day**
 - AM is between midnight (12 am) and midday (12 pm)
 - PM is between midday (12 pm) and midnight (12 am)
- A **24-hour clock** goes around once for **one complete day**
 - The display uses four digits, two for the hour, two for the minutes
 - The day starts at midnight which is 00 00 (12 am) and goes up to 23 59 (11.59 pm)
- You need to be able to **read the time** from both a **12-hour clock** and a **24-hour clock**



Your notes

TIME OF DAY

A NEW DAY STARTS AT MIDNIGHT WHICH IS 12am ON THE 12-HOUR CLOCK AND 00 00 ON THE 24-HOUR CLOCK SYSTEMS.

am	IS	MIDNIGHT	TO	MIDDAY
		12 am		12 pm
		00 00		12 00
pm	IS	MIDDAY	TO	MIDNIGHT
		12 pm		12 am
		12 00		00 00

IGNORING SECONDS, 24-HOUR CLOCKS RUN FROM 00 00 TO 23 59

00 02	IS	12:02 am
09 37	IS	9:37 am
12 10	IS	12:10 pm
17 32	IS	5:32 pm

TO GET THE pm HOURS
ADD OR SUBTRACT 12
TO THE HOURS DIGITS

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How do I read the time from an analogue clock?

- An **analogue** clock works in **12-hour** time
 - The **long hand** is the **hour hand**
 - Each number on the clock represents **one hour** for the hour hand

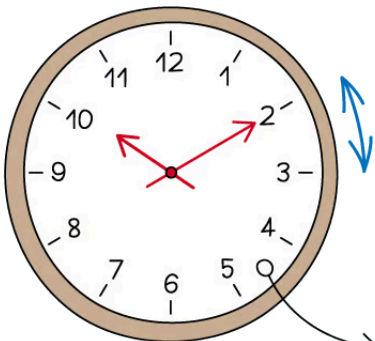
▪ The **short hand** is the **minute hand**

- Each number on the clock represents **five minutes** for the minute hand
- Some clocks will have markings for individual minutes

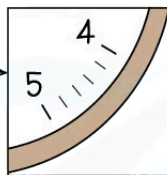


Your notes

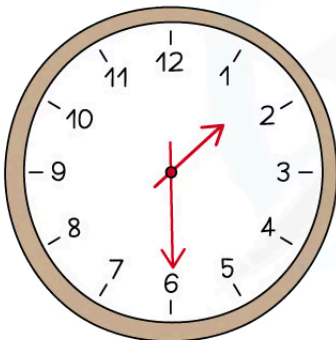
ANALOGUE CLOCKS



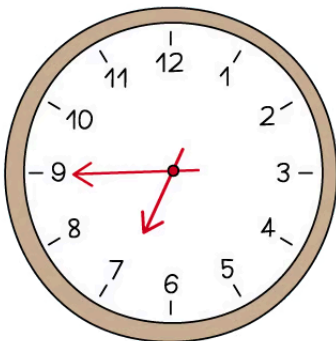
SHOWS 10 10 (OR 22 10)
THERE ARE 5 MINUTES
BETWEEN EACH NUMBER



SOME CLOCKS MAY
HAVE MARKING FOR
EACH MINUTE



01 30 (OR 13 30)
"HALF PAST ONE".
THE HOUR HAND IS
HALFWAY BETWEEN
1 AND 2



06 45 (OR 18 45)
"QUARTER TO SEVEN".
THE HOUR HAND CLOSER
TO 7 THAN 6.

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How do I read the time from a digital clock?

- A **digital** clock can use either **24 hour** time or **12-hour** time
 - A colon ':' is often displayed between the hours and minutes
 - E.g., 1245 would be displayed as 12:45
 - AM or PM does **not** need to be specified with 24-hour time
 - it may or may not be shown on a 12-hour time digital clock
- For **single-digit** hours, clocks often miss out the first zero
 - e.g. 09:23 would be displayed as 9:23



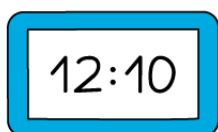
Your notes



Your notes

DIGITAL CLOCKS

e.g. A DIGITAL CLOCK SHOWING 12:10 am
IN 12-HOUR CLOCK MODE WILL SHOW

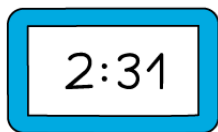


am OR pm
MAY OR MAY NOT
BE SHOWN

IN 24-HOUR CLOCK MODE IT WILL SHOW



e.g. 2 FOR SINGLE-DIGIT HOURS IN 24-HOUR
CLOCK MODE, DIGITAL CLOCKS OFTEN
MISS OUT THE FIRST ZERO.
02 31 WOULD SHOW AS



YOU SHOULD WRITE
THIS WITH FOUR
DIGITS: 02 31

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How do I calculate with time using the 12-hour clock?

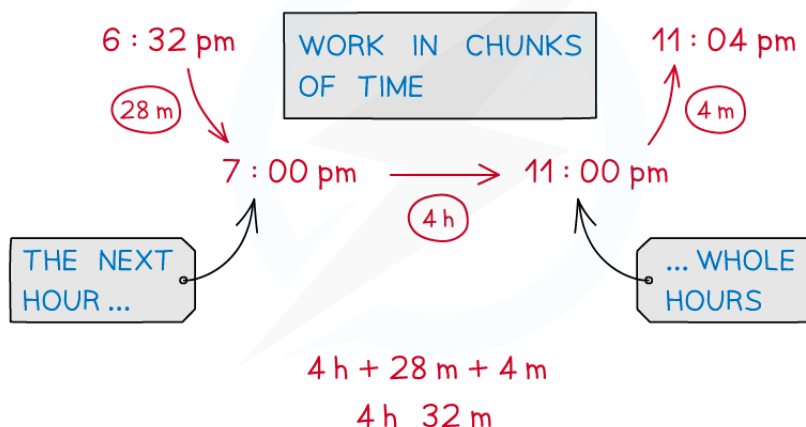
- Work in **chunks** of time
 - Calculate the **minutes until the next hour**, then **whole hours**, then **minutes until a final time**
- Ensure you know when the 12-hour clock **switches** from **AM to PM** and vice versa



Your notes

12-HOUR CLOCK CALCULATION

e.g. HOW LONG IN HOURS AND MINUTES ARE THERE BETWEEN 6:32 pm AND 11:04 pm?



THERE ARE 4 HOURS AND 32 MINUTES BETWEEN 6:32 pm AND 11:04 pm

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How do I calculate with time using the 24-hour clock?

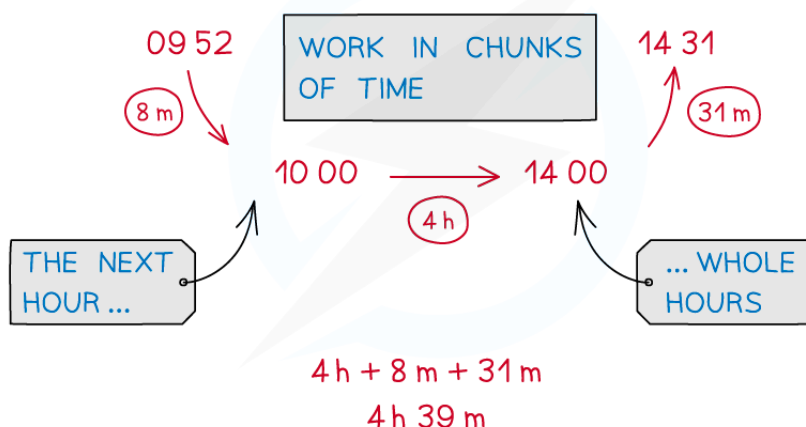
- Work in **chunks** of time just like the 12-hour clock calculations
 - Calculate the **minutes until the next hour**, then **whole hours**, then **minutes until a final time**
- If the hour is greater than 12, subtract 12 from it to find the 12-hour **PM** hour



Your notes

24-HOUR CLOCK CALCULATION

e.g. HOW LONG IN HOURS AND MINUTES ARE THERE BETWEEN 09 52 AND 14 31?



THERE ARE 4 HOURS AND 39 MINUTES BETWEEN 09 52 AND 14 31

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How do I use bus and train timetables?

- Bus and train **timetables** tend to use the **24-hour** clock system
 - Each **column** represents a **different bus or train**
 - Times** are listed as **four digits** without the colon ':'
 - The time in each cell usually indicates the **departure time** (when the bus/train leaves that stop/station)
 - The **last location** on the list usually shows the **arrival time**

BUS TIMETABLE:

CORONATION STREET	0750	0800	0816
ALBERT SQUARE	0818	0830	0849
RAMSEY STREET	0825	0840	0903
EMMERDALE VILLAGE	0834	0852	0918

LOCATION

EACH COLUMN IS
A DIFFERENT BUS

24-HOUR
CLOCK

TIMES SHOWN ARE
DEPARTURE TIMES

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How do I convert time between time zones?

- Different countries are in different **time zones**, depending on where they are on the planet
- For example when it is 11:36 in London in the UK, it may be 06:36 in New York in the USA
 - So New York is 5 hours **behind** London
 - London is 5 hours **ahead** of New York
- You must consider:
 - What is the **time difference** between the two locations (**in hours**)?
 - Which time zone is **ahead**, and which one is **behind**?
- For example, Buenos Aires (Argentina) is 6 hours behind Helsinki (Finland)
 - When it is 7 am in Helsinki
 - $7 \text{ am} - 6 \text{ hours} = 1 \text{ am}$ in Buenos Aires
 - When it is 5 pm in Buenos Aires
 - $5 \text{ pm} + 6 \text{ hours} = 11 \text{ pm}$ in Helsinki



Examiner Tips and Tricks

- Be careful when using a calculator, as they can often cause problems in time-based questions!
 - Remember that 1 hour 30 mins of time would be 1.5 hours on a calculator
- You can use the **degrees, minutes and seconds function**  to enter time on your calculator
 - E.g. 3 minutes 45 seconds would be entered as $0^{\circ}3'45''$
 - You can convert between an answer on the calculator, given in either format of time, by pressing this button



Your notes



Worked Example

(a) A film starts at 15 35 and lasts for 1 hour 45 minutes.

Find the time at which the film ends.

Work in chunks of time

Add the hour on

$$15\ 35 + 1\ \text{hour} = 16\ 35$$

Add on the number of minutes until the next hour

There are 60 minutes in an hour

Add on 25 minutes of the remaining 45 minutes to get to the next hour

$$16\ 35 + 25\ \text{minutes} = 17\ 00$$

There are $45 - 25 = 20$ minutes left of the film to add

Write your final answer using the type of clock used in the question (24-hour) unless otherwise stated

17 20

(b) It is due to start raining at 02 35 and not stop until 08 14.

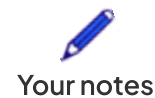
Find the time, in hours and minutes, it is expected to rain for.

Work in chunks of time

Find the number of minutes until the next hour

$$02\ 35 + 25\ \text{minutes} = 03\ 00$$

Find the number of hours until the hour in the stopping time



$$03\ 00 + 5\ \text{hours} = 08\ 00$$

Find the remaining minutes until the rain stops

$$08\ 00 + 14\ \text{minutes} = 08\ 14$$

Add together the hours and minutes

$$25\ \text{minutes} + 5\ \text{hours} + 14\ \text{minutes}$$

5 hours 39 minutes

(c) A train journey started at 11.23 am and finished at 2.27 pm.

How long, in hours and minutes, was the train journey?

Work in chunks of time

Find the number of minutes until the next hour

Be careful about the AM/PM boundary

$$11.23\ \text{am} + 37\ \text{minutes} = 12.00\ \text{pm}$$

Add on the number of hours until the hour in the arrival time

$$12.00\ \text{pm} + 2\ \text{hours} = 2.00\ \text{pm}$$

Add on the remaining minutes until the arrival time

$$2.00 + 27\ \text{minutes} = 2.27\ \text{pm}$$

Add up the hours and minutes to find the total length of the train journey

$$37\ \text{minutes} + 2\ \text{hours} + 27\ \text{minutes}$$

$$37\ \text{minutes} + 27\ \text{minutes add up to 1 hour and 4 minutes}$$

3 hours 4 minutes

(d) The table below shows part of a bus timetable.

Coronation Street	0750	0800	0816
Albert Square	0818	0830	0849
Ramsey Street	0825	0840	0903
Emmerdale Village	0834	0852	0918

How long should the journey between Coronation Street and Ramsey street take on the 0800 bus?

The 08 00 bus is the middle column



Your notes

Find the time between the departure times at Coronation Street and Ramsey street for the middle column

08 40 - 08 00

40 minutes



Worked Example

The time in Madrid is 7 hours ahead of the time in Chicago.

Mark takes a flight from Madrid to Chicago which lasts 9 hours and 45 minutes.

The flight leaves at 7:30 am Madrid time.

What will the local time be in Chicago when he arrives?

Work in Madrid time, and then change at the end

Add 9 hours 45 minutes on to 7:30 am to find the time Mark arrives in Chicago (in Madrid time)

$$07:30 + 9 \text{ h } 45\text{m} = 16:30 + 45\text{m} = 17:15$$

17:15 is the time in Madrid, so find the time in Chicago

Madrid is 7 hours ahead of Chicago

Subtract 7 hours

$$17:15 - 7 \text{ hours} = 10:15$$

10:15 am



Your notes

Money Calculations

Money Calculations

What currencies can be used?

- Many different currencies are used
- The **most commonly** used **currencies** are
 - US Dollars (\$) or USD)
 - Great British Pounds (£ or GBP)
 - Euros (€ or Euros)
 - It is possible to see other currencies used, with or without their symbols

What might I be asked to do in money calculations?

- Read exam questions carefully to identify **key words**
 - **Total** or **sum** will mean to **add up**
 - **Difference** or **increase/decrease** in costs will involve **subtracting**
 - Changing from one currency to another (**exchange rates**) will involve **multiplying or dividing**
- Some questions may involve a **combination** of these
 - E.g., Working out the total cost of an energy bill
- Questions that involve money may also involve **other topics** in the course
 - E.g., Fractions, percentages, simple and compound interest etc.

How should I round values in a money calculation?

- Many currencies will be rounded to **two decimal places**
 - Dollars, pounds and euro should all be used to two decimal places
 - Always write **down both decimal places**, even if the second is zero
 - This is particularly important when using a **calculator**
 - 1.4 dollars on a calculator should be written as \$1.40
- For a question involving **large numbers** such as the cost of a car



Your notes

- Rounding to the nearest dollar, 10 dollars or 100 dollars may be **more appropriate**
 - Use the **information in the question** to make a judgement
- **Some currencies** have large numbers due to exchange rates
 - These are usually rounded to the **nearest whole number**
 - E.g., \$10 is 816.38 rupees making \$100 the same as 8163.80 rupees
 - Rounding the exchange rate to \$100 is 8160 rupees would be appropriate

What should I do when money calculations involve more than two decimal places?

- In some contexts money facts may be given to **more than two decimal places**
 - E.g. One litre of petrol in the UK costs an average price of £1.579
- Use **all of the decimal places** given in your **working**
 - Round (to two decimal places or whatever is appropriate) for your **final answer only**



Examiner Tips and Tricks

- Use the **information given in the question** to decide how to **round** your final answer
- Check that your answer **matches** the currency in the question



Worked Example

In his favourite UK fashion store, Thomas buys 4 t-shirts costing £8.50 each and 2 pairs of shorts costing £7.20 each. On his way home Thomas fills his car up with 45 litres of petrol at a price of £1.579 per litre.

Find out how much Thomas spent in total on clothes and petrol.

Find the total cost of the t-shirts, shorts and petrol separately
Use the figures as they are given, do not round any at this stage

$$4 \times 8.50 = 34$$

$$2 \times 7.20 = 14.4$$

$$45 \times 1.579 = 71.055$$

Total (add) these amounts

$$34 + 14.4 + 71.055 = 119.455$$

The currency is Great British Pounds (£) and values are relatively small so it makes sense to round the final answer to two decimal places

Thomas spends a total of £119.46 on clothes and petrol



Your notes



Your notes

Exchange Rates

Exchange Rates

Exchange rates are used to **compare** and **convert** between **different currencies**.

How do I convert between currencies?

- Start by writing the **exchange rate**
 - £1.00 (GBP) = €1.17 (EUR)
- To convert **from GBP to EUR**, **multiply** by 1.17
 - Because $1.00 \times 1.17 = 1.17$
- To convert **from EUR to GBP**, **divide** by 1.17
 - Because $1.17 \div 1.17 = 1.00$
- To find £28 in Euros
 - $28 \times 1.17 = €32.76$
- To find €75 in Pounds
 - $75 \div 1.17 = £64.10$ (to the nearest penny)



Worked Example

On a particular day, the exchange rate from US Dollars to Japanese Yen is:

$$1 \text{ US Dollar} = 155.62 \text{ Japanese Yen}$$

and the exchange rate from Australian Dollars to Japanese Yen is:

$$1 \text{ Australian Dollar} = 102.09 \text{ Japanese Yen}$$

An item costs **40 US Dollars**.

Find the cost of the item in **Australian Dollars**.

The first exchange rate is $1 \text{ US Dollar} = 155.62 \text{ Japanese Yen}$

So to convert from US Dollars to Japanese Yen, multiply by 155.62

$$40 \times 155.62 = 6224.80 \text{ Japanese Yen}$$

The second exchange rate is 1 Australian Dollar = 102.09 Japanese Yen
So to convert from Japanese Yen to Australian Dollars, divide by 102.09

$$6224.80 \div 102.09 = 60.9736... \text{ Australian Dollars}$$

Round to 2 decimal places as we are dealing with money

60.97 Australian Dollars



Your notes